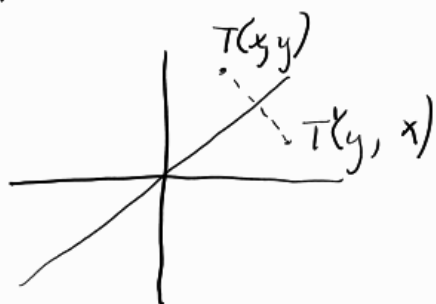


(7.)

Poišči matriko za linearno preslikavo

a) zrcaljenje in $pr_{\perp}$ na premico $y=x$
 $z(x,y) = (y,x)$



$$z \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ x \end{bmatrix}$$

$$z = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$z^2 = I$$

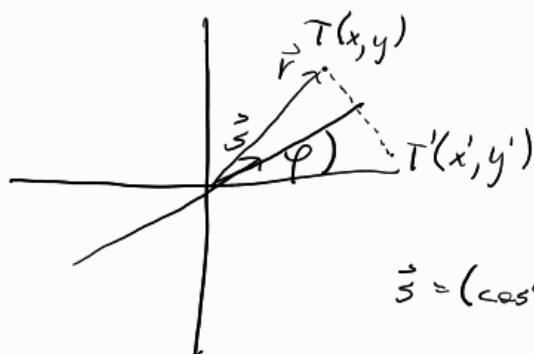
$$P(x,y) = \left(\frac{x+y}{2}, \frac{x+y}{2} \right)$$

$$\begin{bmatrix} \frac{x+y}{2} \\ \frac{x+y}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

" P

$$\boxed{P^2 = P} \text{ - velja za projekcije}$$
$$P^T = P$$

b) zrcaljenje in $pr_{\perp}$ na $\vec{r}(t) = t(\cos\varphi, \sin\varphi)$, $\varphi \in (0, 2\pi)$



$$P_{\varphi}(x,y) = (\vec{r} \cdot \vec{s}) \cdot \vec{s} = (x \cos\varphi + y \sin\varphi)(\cos\varphi, \sin\varphi)$$

$$\vec{s} = (\cos\varphi, \sin\varphi)$$

$$\vec{r} \cdot \vec{s} = x \cos\varphi + y \sin\varphi$$

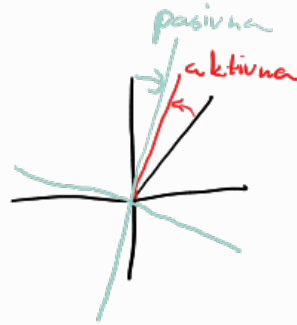
$$P_{\varphi} = \begin{bmatrix} \cos^2\varphi & \cos\varphi\sin\varphi \\ \cos\varphi\sin\varphi & \sin^2\varphi \end{bmatrix}$$

$$Z_\varphi(x, y) = (x, y) + 2 \cdot (P_\varphi(x, y) - (x, y)) = 2P_\varphi(x, y) - (x, y) = (2x \cos^2 \varphi + 2y \cos \varphi \sin \varphi - x, 2x \cos \varphi \sin \varphi + 2y \sin^2 \varphi - y)$$

$$Z_\varphi = \begin{bmatrix} 2\cos^2 \varphi - 1 & 2\cos \varphi \sin \varphi \\ 2\cos \varphi \sin \varphi & 2\sin^2 \varphi - 1 \end{bmatrix} = \underline{\underline{\begin{bmatrix} \cos 2\varphi & \sin 2\varphi \\ \sin 2\varphi & -\cos 2\varphi \end{bmatrix}}}$$

Lin. pr. med $\mathbb{R}^2$
1) rotacija

$$R_\varphi = \begin{bmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{bmatrix} \rightarrow \text{okoli izhodišča}$$



$$R_\varphi^T \cdot R_\varphi = I \rightarrow \text{ortogonalna}$$

↓ dolžina stolpcev je 1
- paroma so $\perp$

$$\det R_\varphi = 1$$

$$\text{Tvori jo grupo } SO(2) = \{R \in \mathbb{R}^{2 \times 2}; R R^T = I, \det R = 1\}$$

2) zrcaljenje

- čez premico $t(\cos \varphi, \sin \varphi)$

$$Z_\varphi = \begin{bmatrix} \cos 2\varphi & \sin 2\varphi \\ \sin 2\varphi & -\cos 2\varphi \end{bmatrix}$$

ortogonalne, $\det Z_\varphi = -1$

ne tvorijo grupe

$$\Rightarrow Z_{\varphi_2} \circ Z_{\varphi_1} = R_{2(\varphi_2 - \varphi_1)}$$

Unija zrc. in rot. tvori grupo

$$O(2) = \{Q \in \mathbb{R}^{2 \times 2}; Q^T Q = I\}$$

3) pr₁

$$P_\varphi = \begin{bmatrix} \cos^2 \varphi & \sin \varphi \cos \varphi \\ \sin \varphi \cos \varphi & \sin^2 \varphi \end{bmatrix}$$

- oblike $S S^T$; S enotski smerni vektor premice

- natanko vse matrike $rk=1$

4) Razteg

$$D_{a,b} = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \rightarrow \text{razteg v } \mathbb{R}^2 \text{ za } a \text{ v smeri } x \\ b \text{ v smeri } y$$

Natančno vse simetrične in pozitivno definitne matrice

$\downarrow$
 M

$\downarrow$

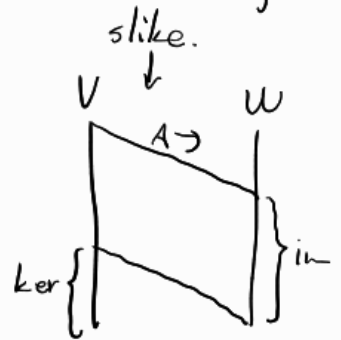
$$\text{za } \forall z \neq 0: z^T M z > 0$$

8. zapiši matriko za projt na ravnino $x+2y+2z=0$ v $\mathbb{R}^3$. Poišči bazo in jedro slike.

$$P(\vec{r}) = \vec{r} - (\vec{r} \cdot \vec{n}) \cdot \vec{n} = \\ = (I - n n^T) \vec{r}$$

$$\vec{n} = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3} \right)$$

$$P = \begin{bmatrix} \frac{8}{9} & -\frac{2}{9} & -\frac{2}{9} \\ -\frac{2}{9} & \frac{5}{9} & -\frac{4}{9} \\ -\frac{2}{9} & -\frac{4}{9} & \frac{5}{9} \end{bmatrix} \sim \begin{bmatrix} 2 & -5 & 4 \\ 4 & -1 & -1 \\ -2 & -4 & 5 \end{bmatrix} \sim \begin{bmatrix} 2 & -5 & 4 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 2 & -5 & 4 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$



$$\ker A = A^{-1}(\{0\}) \subseteq \mathbb{R}^n \\ \text{im } A = A(\mathbb{R}^n) \subseteq \mathbb{R}^m$$

$$\rightarrow \text{rk}(P) = 2 = \dim(\text{im } P)$$

$$\dim(\ker P) = 1$$

$$\text{im}(P) = \text{Span}\{(4, -1, -1), (-2, 5, -4)\}$$

$\ker(P)$:

$$\begin{bmatrix} 2 & -5 & 4 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$2x - 5y + 4z = 0 \quad 2x = z \quad \vec{r} = z \left(\frac{1}{2}, 1, 1 \right)$$

$$y - z = 0 \rightarrow y = z$$

$$\ker(P) = \text{Span}\{(1, 2, 2)\}$$

9. $\vec{w} = (w_x, w_y, w_z) \in \mathbb{R}^3$

$$W: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$W(\vec{r}) = \vec{w} \times \vec{r}$$

zapiši matriko W in poišči bazo jedra in slike za $\vec{w} = (1, 1, 1)$

$$\begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} w_y z - w_z y \\ w_z x - w_x z \\ w_x y - w_y x \end{bmatrix} \rightarrow W = \begin{bmatrix} 0 & -w_z & w_y \\ w_z & 0 & -w_x \\ -w_y & w_x & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 1 \\ 0 & -1 & 1 \end{bmatrix} \rightarrow \text{Rk}(W) = 2 = \dim(\text{im} W)$$

$$\dim(\text{Ker} W) = 1$$

$$\text{im}(W) = \text{Span}\{(0, 1, -1), (-1, 0, 1)\}$$

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$x - z = 0 \rightarrow x = z$$

$$z - y = 0 \rightarrow z = y \rightarrow x = y = z$$

$$\text{Ker}(W) = \text{Span}\{(1, 1, 1)\}$$

10. $T: \mathbb{R}^{2 \times 2} \rightarrow \mathbb{R}^{2 \times 2}$

$$T(X) = AX - XA = [A, X] ; A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

Pokaži, da T linearna, izračunaj matriko, poišči bazo jedra in slike.

- homogena $T(\alpha X) = \alpha T(X)$

- aditivna $T(X+Y) = T(X) + T(Y)$

$\Rightarrow T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ je linearna, če so komponente lin. funkc. več spt.

BREZ PROSTIH ČLENOV

homogenost:

$$T(\alpha X) = A\alpha X - \alpha XA = \alpha T(X) \checkmark$$

aditivnost

$\Rightarrow T$ je linearna

$$T(X+Y) = A(X+Y) - (X+Y)A = AX + AY - XA - YA = T(X) + T(Y) \checkmark$$

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} - \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} a+c & b+d \\ c-a & d-b \end{bmatrix} - \begin{bmatrix} a-b & a+b \\ c-d & c+d \end{bmatrix} = \underline{\underline{\begin{bmatrix} b+c & d-a \\ d-a & b-c \end{bmatrix}}}$$

Baza:

$$\mathcal{B} = \left(\overset{B_1}{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}}, \overset{B_2}{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}, \overset{B_3}{\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}}, \overset{B_4}{\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}} \right)$$

$$T\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right) = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} = 0B_1 - B_2 - B_3 + 0B_4$$

$$T\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = B_1 - B_4$$

$$T\left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = B_1 - B_4$$

$$T\left(\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = B_2 + B_3$$

$$\rightarrow J = \begin{bmatrix} 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & -1 & 0 \end{bmatrix}$$

matrika lin. presl.

↓

$$T\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right) = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ -1 \\ 0 \end{bmatrix}$$

11.

$$\mathcal{E} = \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

$$\mathcal{B} = \left(\begin{bmatrix} \cos \varphi \\ \sin \varphi \end{bmatrix}, \begin{bmatrix} -\sin \varphi \\ \cos \varphi \end{bmatrix} \right), \quad \varphi \in [0, 2\pi]$$

Dokazi prehodno matriko iz baze $\mathcal{E}$ v $\mathcal{B}$.

$$\begin{bmatrix} \cos \varphi \\ \sin \varphi \end{bmatrix} = \cos \varphi \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \sin \varphi \begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow P_{\mathcal{B}\mathcal{E}} = \begin{bmatrix} \cos \varphi & \sin \varphi \\ -\sin \varphi & \cos \varphi \end{bmatrix}$$

$$\begin{bmatrix} -\sin \varphi \\ \cos \varphi \end{bmatrix} = -\sin \varphi \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \cos \varphi \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$P_{\mathcal{E}\mathcal{B}} = P_{\mathcal{B}\mathcal{E}}^{-1} = \begin{bmatrix} \cos \varphi & \sin \varphi \\ -\sin \varphi & \cos \varphi \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \frac{1}{\det} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$